

Consumption-related emotions over time: Fit between prediction and experience

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Abstract Consumers base many decisions on affective forecasts, i.e., predictions about future emotions. These predictions, however, are susceptible to systematic biases. The present paper examines the accuracy of predicting emotional change in the consumer context. The goals of the study were to examine whether predictions of a change in consumption-related emotions fit actual experienced change and whether original predictions are remembered correctly. A two-wave longitudinal survey with 86 consumers in Austria and the UK showed that consumers who had bought shoes overestimated the decrease of positive consumption-related emotions over time. Results indicate that the intuitive theory of adaptation is related to the prediction bias. Moreover, participants misremembered original predictions as more consistent with actual experienced emotions. This hindsight bias hinders consumers from learning from previous prediction errors.

Keywords Affective forecasting · Prediction bias · Hindsight bias · Consumption-related emotions · Consumer behavior

When making decisions, people often take into account how different options will make them feel and how long these feelings will last, and they consider these predicted emotions in their decisions (Loewenstein et al. 2001; Mellers et al. 1999;

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Shiv and Huber 2000; Simonson 1992). For example, consumers who are planning to buy a new car may picture driving the car and may predict positive emotions associated with this experience. Consumers may also predict how these emotions will change with the duration of ownership—whether emotions last for a long time or fade away quickly. These predictions about change in emotions may then influence the decision to buy the car or the amount of money consumers are willing to spend. When thinking about the change in emotions, consumers may take into account beliefs about adaptation—how quickly they will get used to the new car. If consumers believe that the car becomes ordinary with time, they will predict decreasing positive emotions associated with the car. However, if this belief about adaptation is not accurate, predictions about decreasing emotions will be incorrect as well.

In the domain of consumer behavior, consumption-related emotions are a relevant factor for decisions (Derbaix and Pham 1991; Havlena and Holbrook 1986; Hirschman and Morris 1982; Holbrook and Hirschman 1982). Consumption-related emotions refer to the set of emotions experienced in the post-purchase period resulting from consumption itself. It was shown that consumption-related emotions relate to consumer satisfaction (Mano and Oliver 1993; Phillips and Baumgartner 2002; Westbrook and Oliver 1991) and post-consumption behavior such as positive and negative word-of-mouth, repurchase intention, and complaint behavior (Ladhari 2007; Soscia 2007). Moreover, not only experienced but also predicted consumption-related emotions and the discrepancy between predicted and experienced emotions influence consumer satisfaction (Phillips and Baumgartner 2002). Consumers should therefore be able to predict consumption-related emotions accurately in order to make optimal decisions and to enhance satisfaction. As many products are consumed over a long period of time, it is particularly important for consumers not only to consider initial pleasure but also pleasure over the total lifetime of the product; therefore, consumers need to consider how consumption-related emotions change the longer they own a product. The main goal of the present study was to compare predictions of a change in consumption-related emotions with actual experienced change in a longitudinal field setting and to explore the role of beliefs about adaptation and of temporal separation. In addition, it was explored whether consumers are able to recall predictions of consumption-related emotions correctly. A better understanding of the dynamics of change in consumption-related emotions and their subjective perception can contribute to better-informed consumer decisions.

1 Prediction bias about change in emotions

Research on *affective forecasting*, i.e., predictions about future emotions, indicates that people make systematic errors in predicting emotions (for reviews, see Loewenstein and Schkade 1999; MacInnis et al. 2005; Wilson and Gilbert 2003). According to MacInnis et al. (2005), the term affect is used relatively broadly in the literature on affective forecasting, including different affective experiences such as visceral feelings, preferences or tastes, generalized valenced feelings, as well as specific emotional states. One stream of research showed that people often overestimate the intensity and duration of emotional reactions to events and, hence, overestimate the impact that future events will have on their general well-being. This so-called *impact bias* has

been documented for positive events such as sports victories (Meyvis et al. 2007; Wilson et al. 2000) or positive test results (Mellers and McGraw 2001), as well as negative events such as romantic breakups (Gilbert et al. 1998) or electoral defeats (Gilbert et al. 1998; Meyvis et al. 2007). Another stream of research, situated in the consumer context, focused not on general feelings after events but on more specific emotions and preferences regarding specific products. Table 1 presents the key findings of these studies. The first three studies examined food choices as index for hedonic predictions. Consumer choices were influenced by the current state of hunger (Gilbert et al. 2002; Read and van Leeuwen 1998), and more variety was chosen when making combined choices for future consumption than when making separate choices immediately before consumption (Simonson 1990). Loewenstein and Adler (1995) examined predicted and actual selling prices for coffee mugs and found that participants were not aware of the endowment effect, underestimating how much they would value an object after receiving it. Two studies showed effects of mispredictions on product evaluation and satisfaction (Patrick et al. 2007; Phillips and Baumgartner 2002). Of special interest for the present study are two studies dealing with predictions of change (Kahneman and Snell 1992; Wang et al. 2009). Kahneman and Snell (1992) examined predictions of a taste change. In a pilot study, participants had to consume ice cream on eight consecutive days. At the beginning of the experiment, they predicted how they would want and like the ice cream over time and then they rated their actual experience every day. Participants expected a decline in liking; however, the actual decrease was smaller than the predicted decrease. In the main study, participants had to consume plain yogurt in a similar setting. Participants again expected a decrease in liking; however, contrary to expectations, their liking for the yogurt increased over time. Although the results of the pilot study and the main study are not completely consistent, one showing a gradual error in the degree of taste change, and the other an error in the direction of taste change, the main conclusion was that people are not able to predict changes in taste. Wang et al. (2009) also examined predictions for different points in time and showed that by making predictions for both near and distant points in the future, a belief in adaptation was cued, and participants thus predicted lower emotions than when only making predictions for one point in time. The main conclusion was that consumers know about adaptation but often fail to take this knowledge into account. This overview of studies in the consumer context shows that there is a lack of field studies dealing with real consumer purchases and the change in consumption-related emotions. Both studies dealing with predictions of change were conducted in a laboratory setting, one using mainly scenarios (Wang et al. 2009) and the other examining enjoyment of food items given to the participants in the course of the experiment (Kahneman and Snell 1992). To derive practical implications for marketing, it seems therefore important to examine predictions in a real-life setting with consumers making real purchases, choosing to spend their own money on a product that is used repeatedly.

A better understanding of consumers' predictions of emotions is important for marketing, both in the pre-purchase and the post-purchase phase. (1) Predictions are relevant in the pre-purchase phase because they can guide consumer decisions (e.g., Shiv and Huber 2000). As marketing stimuli can evoke emotional predictions, certain marketing strategies for hedonic products, e.g., communicating long-lasting pleasure with holidays or luxury products, could be used to influence consumer decisions.

Table 1 Summary of relevant studies dealing with predictions in the consumer context

Study	Participants	Domain/ product	Measures	Time horizon	Results
Gilbert et al. (2002)	S 1: 61 students	S 1: scenario spaghetti	S 1: predicted food enjoyment	S 1: next morning and next evening	Predictions of food enjoyment and food purchases influenced by current hunger
	S 2: 135 consumers	S 2: food purchases	S 2: intended and actual food purchases	S 2: same day	
Read and van Leeuwen (1998)	200 employees	Healthy vs unhealthy snacks	Advanced and immediate choices	1 week	Choices for future consumption influenced by current hunger
Simonson (1990)	S 1: 67 students	S 1: scenario groceries	S 1: hypothetical choice	S 1: 3 days	Diversification bias: more variety chosen when combined choices for future consumption than separate choices immediately before consumption
	S 2: 392 students	S 2 and 3: snacks	S 2 and 3: actual choice	S 2 and 3: 3 weeks (one measurement per week)	
	S 3: 46 students				
Loewenstein and Adler (1995)	S 1: 27 students, 42 adults	Coffee mug	Predicted and actual value (selling price)	Same day	Unawareness of endowment effect: underestimation of valuing an object after receiving it
	S 2: 106 students				
Patrick et al. (2007)	S 1: no information	S 1: film clip	Product evaluation	One measurement	Product evaluation only influenced by errors in affective forecasting when experience worse than forecasted
	S 2: 99 students	S 2: piece of music			
Phillips and Baumgartner (2002)	S 1: 53 students	Orange juice	Predicted and actual consumption emotions	Same day	Satisfaction influenced by affective predictions, experience, and disconfirmation
	S 2: 51 students				
Kahneman and Snell (1992)	Pilot S: 16 adults	Pilot S: ice cream	Predicted and actual liking over time	Eight consecutive days	Pilot S: actual decline in liking smaller than predicted decline
	Main S: 154 students	Main S: plain yogurt			Main S: increase in liking, decline in liking expected
Wang et al. (2009)	S 1: 175 students	S 1: kaleidoscope	Predicted enjoyment (in study 1 also actual enjoyment)	S 1: 1 day, 1 week	Beliefs about adaptation cued and predictions adjusted when making predictions for different points in time. Adaptation ignored when making predictions for one point in time. Purchase decisions influenced by the cued belief in adaptation
	S 2: 193 students	S 2: scenario new car		S 2: immediately, several months	
	S 3: 89 students	S 3: scenario kaleidoscope		S 3: 1 week, 1 day	
	S 4: 85 students	S 4: scenario digital camera		S 4: 1 year, 1 week, 2 weeks	
	S 5: 157 students	S 5: scenario ipod nano		S 5: 1 week, 2 years	

S study

Furthermore, emotional predictions could explain why consumers spend large sums on hedonic products and why consumers prefer well-known brands over no-name brands as they might associate longer lasting pleasure with the former. (2) The accuracy of such predictions is relevant in the post-purchase phase because mispredictions can lead to consumer dissatisfaction and reduced brand loyalty (e.g., MacInnis et al. 2005; Phillips and Baumgartner 2002). Consumer errors in predictions can become costly: Over- as well as underestimation of consumption-related emotions is unfavorable for both consumers and the economy. If consumers err in the direction of overestimating the intensity and stability (i.e., the degree of change over time) of positive consumption-related emotions, they are prone to overconsumption, i.e., spending money on products and services that do not provide as much positive emotions as expected. Not only could this money be used more efficiently, such errors also lead to disappointment and dissatisfaction and could contribute to consumer indebtedness. If consumers underestimate positive emotions, they are prone to underconsumption, i.e., missing opportunities to gain pleasure from consumption. For negative consumption-related emotions, overestimation would contribute to underconsumption and underestimation to overconsumption. Therefore, it is advisable that marketers aim at making predictions more accurate, both in the initial level of consumption-related emotions and in the change of emotions over time.

Furthermore, it is of interest for consumers and marketers to not only examine if consumers are able to predict a change in consumption-related emotions accurately but also to identify the sources of these errors in order to find ways to reduce a potential bias. Several sources of errors in predicting emotions are discussed in the literature (for reviews, see Loewenstein and Schkade 1999; MacInnis et al. 2005; Wilson and Gilbert 2003). Most explanations however focus on the emotional consequences of single events. As the present study focuses on changes of emotions, two explanations dealing with changes over time are of particular relevance. The first explanation relates to incorrect *intuitive theories* of hedonics. People hold theories about what makes them happy or unhappy and also theories about changes of emotion, of adaptation, or sensitization (Gilbert et al. 1998; Loewenstein and Schkade 1999; Wilson and Gilbert 2003). When it comes to predicting their own tastes and emotions, people often use these intuitive theories (Igou 2004). If the theories do not correspond with reality, prediction errors occur. As different theories (e.g., theory of adaptation vs theory of sensitization) lead to different predictions, depending on the kind of theory, they can produce over- as well as underestimation of future emotions (Wilson and Gilbert 2003). When examining predictions about changes in emotions, the intuitive theory of adaptation should be taken into account. There is some evidence that people tend to overestimate adaptation processes (Hoelzl et al. 2009; Kahneman and Snell 1992; Snell et al. 1995). However, Wang et al. (2009) highlight that although people may be aware of adaptation, they often do not consider this knowledge in their predictions. The second explanation argues that prediction errors occur due to the *temporal separation* between prediction and actual experience and that errors especially arise when this time period is long (MacInnis et al. 2005). When predicting emotional reactions to events, people fail to consider the temporal context of the event, and this failure is more pronounced for distant events. A larger time period between prediction and actual experience therefore can be assumed to increase the magnitude of the prediction error.

2 Perceived prediction bias after experience

For repeated experiences, it could be expected that consumers learn from their mistakes. However, prediction errors seem to persist even for frequent events. One possible explanation why people fail to learn from previous prediction errors is the *hindsight bias* (for reviews, see Christensen-Szalanski and Willham 1991; Hawkins and Hastie 1990). One form of the hindsight bias or the “I-knew-it-all-along effect” (Wood 1978) refers to the tendency of people to change their judgment of past events: Once they know the outcome of the event, they believe that they would have predicted the actual outcome. This form of hindsight bias is a stable phenomenon found in several settings related to forecasting, for instance, the probability of historical events (Fischhoff 1975) or economic developments (Hoelzl et al. 2002). Of particular relevance for the present study, the hindsight bias was also found for expectations of product performance (Zwick et al. 1995) and predictions of emotional reactions (Meyvis et al. 2007). Zwick et al. (1995) showed that experience with a product biases people’s recall of their pre-purchase expectations in the direction of perceived product performance. Furthermore, recalled predictions of product performance had more influence on consumer satisfaction than original predictions. The hindsight bias implies that people do not realize the extent of their prediction errors as they think that they had anticipated the actual outcome correctly and therefore fail to learn from their experience. Meyvis et al. (2007) showed that people only learned from previous prediction errors when they were shown their original predictions or when they were able to recall their predictions accurately. The failure to learn from previous prediction errors highlights the importance of taking into account the hindsight bias when examining prediction biases.

To summarize, the present study examines predictions, experience, and recollections of consumption-related emotions. Based on the literature on affective forecasting, it is expected that a prediction bias will occur, and consumers are incorrect in predicting the change in consumption-related emotions over time. Two possible process explanations of prediction biases dealing with changes over time are explored: (a) intuitive theories about adaptation would suggest that such beliefs can be linked to the prediction bias, and (b) temporal separation would suggest that the temporal distance between prediction and experience influences the magnitude of the prediction bias. Based on the literature on hindsight bias, it is expected that consumers misperceive the degree of their prediction bias. We study these questions in a longitudinal field study with consumers making actual purchases.

3 Method

3.1 Material and procedure

A two-wave longitudinal survey was conducted in two countries: Austria (A) and the UK. The first wave was carried out in shoe shops with consumers who had just bought a pair of shoes. Shoes were chosen as the product because they were shown to have not only utilitarian but also hedonic value (Voss et al. 2003) and are frequently bought

products. This fact is illustrated by the size of the European footwear consumer market, which mounted to almost 3 billion pairs of shoes in 2007. The apparent footwear consumption in the European Union presented a growth of approximately 30% between 2004 and 2007 (European Commission 2008).

Participants had to rate current consumption-related emotions right after the purchase ('current emotions at wave one'). They also had to predict how they would feel about their shoes at a later point in time, either 2 or 4 weeks later ('predicted emotions'). Participants were randomly assigned to either the 2-week or the 4-week condition. The time intervals of 2 and 4 weeks were chosen because they are clearly distinguishable and at the same time assessable. At the end of the questionnaire, an email or postal address was solicited, and for the second wave, which was conducted 2 (4) weeks later, a follow-up questionnaire was sent to the participants. At the second wave, participants again had to rate their current emotions about that particular pair of shoes ('current emotions at wave two') and were asked to recall their original predictions ('recalled predictions'). When answering the second questionnaire, participants had the chance to win vouchers from their shoe shop.

Consumption-related emotions Participants were asked how they felt regarding the particular pair of shoes, using selected items of the Positive and Negative Affect Scale (PANAS; Watson et al. 1988), including positive (proud, excited, enthusiastic, interested, and inspired) as well as negative emotions (nervous, afraid, guilty, distressed, and irritable), measured on a 5-point scale (ranging from 'not at all' to 'extremely'). The PANAS Scale was chosen as it is a reliable, valid, and stable measure for emotions. Stability over time is especially important for the present study as it is dealing with emotions at different points in time. There is a large degree of overlap between the ten items we selected with, e.g., the consumption emotions set from Richins (1997). Due to floor effects in all negative emotions ($M=1.02-1.23$, $SD=0.21-0.81$), analyses were restricted to positive emotions. The mean of the five items on positive emotions was used as a scale for (a) current emotions at wave one (Cronbach's $\alpha=0.81$), (b) predicted emotions at wave one (Cronbach's $\alpha=0.88$), (c) current emotions at wave two (Cronbach's $\alpha=0.86$), and (d) recalled predictions at wave two (Cronbach's $\alpha=0.90$).

Adaptation In order to analyze the role of adaptation processes, at the first wave, participants predicted adaptation by reporting to what extent they believed they would get used to the shoes in the following 2 (4) weeks (9-point scale, ranging from 'not at all' to 'completely'). At the second wave, they reported actual experienced adaptation on the same scale.

Demographic and control variables At the first wave, the demographic variables age, gender, and education were collected. Furthermore, the questionnaire of the first wave included control items, such as how many pairs of shoes participants bought per year, whether the particular purchase was planned or unplanned, the price of the shoes they had purchased, and how they were feeling on the day of the survey (5-point scale, ranging from 'very good' to 'very bad'). At the second wave, they again reported how they felt at the day of the survey.

3.2 Participants

At the first wave, 179 participants were recruited; 86 participants answered the follow-up questionnaire. From these, 39 were in the 2-week and 47 in the 4-week condition. Sixty-five participants were female and 21, male. Age ranged from 15 to 76 with a median of 30.5 years. More than half of the participants (48) had a university degree.

Two of the demographic or control variables were found to differ between experimental conditions or countries: Austrian participants ($Md=26$, $M=29.15$, $SD=9.96$) were younger than UK participants ($Md=44$, $M=43.49$, $SD=16.90$), Mann–Whitney $Z=-3.67$, $p<.01$. Participants of the 2-week condition felt better on the day of wave two ($Md=4.00$, $M=4.15$, $SD=0.85$) than participants of the 4-week condition ($Md=4.00$, $M=3.77$, $SD=0.94$), Mann–Whitney $Z=-1.99$, $p=.049$.

A non-responder analysis showed no significant differences between participants who did and did not respond at wave two in the relevant variables such as current emotions, predicted emotions, demographic, and control variables as measured at wave one.

4 Results

To analyze the prediction bias and the hindsight bias, all four ratings of emotions were compared, including the two experimental conditions (2 vs 4 weeks) and the two countries (Austria vs UK). A 4 (time: current emotions wave one, predicted emotions, current emotions wave two, recalled predictions, within subjects) $\times$ 2 (group: 2 vs 4 weeks, between subjects) $\times$ 2 (country: A vs UK, between subjects) analysis of variance¹ revealed a significant effect of time $F(3, 246)=9.97$, $p<.01$. Neither the group effect, $F(1, 82)=3.20$, $p=.08$, nor the country effect, $F(1, 82)=0.43$, $p=.51$, nor any of the interactions were significant (group $\times$ country: $F(1, 82)=0.03$, $p=.86$; group $\times$ time: $F(3, 246)=0.17$, $p=.91$; country $\times$ time: $F(3, 246)=0.52$, $p=.67$; group $\times$ country $\times$ time: $F(3, 246)=0.49$, $p=.69$). Descriptive statistics are shown in Table 2; Fig. 1 illustrates the pattern. Additional analyses showed that this pattern remains unchanged considering the control variables price of the shoes, frequency of buying shoes, and planning as between-subject factors. Apart from the significant time effect, $F(3, 234)=10.89$, $p<.01$, only the main effect of frequency of buying shoes reached significance, $F(1, 78)=6.68$, $p=.01$. Participants who frequently buy shoes (more than three pairs per year) generally indicated more positive consumption-related emotions than participants who buy shoes less frequently (one to three pairs per year). However, the non-significant interactions indicate that the differences between the four ratings of emotions remain unchanged

¹ An ANCOVA was also conducted, including the variables age and general feeling at wave two as covariates, as these variables were found to differ between conditions or countries. The analysis revealed the same results with a significant time effect, $F(3, 246)=9.97$, $p<.01$, a non-significant group effect, $F(1, 80)=2.37$, $p=.13$, a non-significant country effect, $F(1, 80)=0.93$, $p=.34$, and non-significant interactions (group $\times$ country: $F(1, 80)=0.03$, $p=.86$; group $\times$ time: $F(3, 246)=0.17$, $p=.91$; country $\times$ time: $F(3, 246)=0.52$, $p=.67$; group $\times$ country $\times$ time: $F(3, 246)=0.49$, $p=.69$). Neither age ($\beta=-0.11$, $p=.38$) nor general feeling at wave two ($\beta=0.09$, $p=.41$) had a significant effect.

for frequent versus infrequent shoe buying, for cheap versus expensive shoes, and planned versus unplanned purchases.

4.1 Prediction bias about change in emotions

To analyze the prediction bias, planned contrasts between current emotions at wave one and predictions and current emotions at wave two were conducted. Analyses showed a significant difference between current emotions at wave one ($M=2.80$) and predictions ($M=2.32$), $F(1, 82)=34.75$, $p<.01$. The effect size for this difference is $d=0.49$ (Dunlap et al. 1996, Formula 3), which can be considered a medium effect (Cohen 1992). Participants therefore predicted that their emotions would decline. Current emotions indeed declined from wave one ($M=2.80$) to wave two ($M=2.56$), $F(1, 82)=4.98$, $p=.03$, $d=.25$. However, predictions ($M=2.32$) were even lower than current emotions at wave two ($M=2.56$), $F(1, 82)=5.37$, $p=.02$, $d=0.24$. This effect, although small, shows that a prediction bias occurred; participants overestimated the decrease of positive emotions over time and therefore underestimated emotions at wave two². All ratings of emotions were significantly correlated, with the correlation between current emotions at wave one and predictions ($r=.73$) being higher than between current emotions at wave one and current emotions at wave two ($r=.55$) and between predictions and current emotions at wave two ($r=.49$).

In the next step, it was analyzed whether the prediction of adaptation plays a role in the predicted decline of emotions. The difference between current emotions at wave one and predictions was calculated as an index for the predicted change in emotions ($M=-0.40$, $SD=0.67$), with negative values indicating lower predictions than current emotions, i.e., a predicted decline of emotions. This predicted change in emotions was significantly correlated with the prediction of adaptation ($r=-.32$, $p<.01$). Participants predicting a decline of emotions also predicted that they would get used to the shoes strongly.

In addition to the relevance of predicted adaptation for the predicted decline of emotions, it was analyzed whether mispredictions of adaptation can be linked to mispredictions of consumption-related emotions. The difference between predictions and current emotions at wave two was calculated as an index for the prediction bias ($M=0.24$, $SD=0.95$), with positive values indicating higher current emotions than prediction, i.e., an underestimation of emotions. Similarly, the difference between predicted adaptation and experienced adaptation was calculated as an index for the misprediction of adaptation ($M=0.45$, $SD=2.30$), with positive values indicating higher experienced adaptation than predicted adaptation, i.e., an underestimation of adaptation. The prediction bias was significantly correlated with the misprediction of adaptation ($r=-.28$, $p=.01$). Participants underestimating emotions at wave two overestimated how strongly they would get used to the shoes, and participants overestimating emotions at wave two underestimated how strongly they would get used to the shoes. Although the predicted level of adaptation ($M=2.86$, $SD=2.34$) was not significantly different from actual experienced adaptation, ($M=3.22$, $SD=2.58$), $t(85)=-1.38$, $p=.17$, these correlations support the theorized role of adaptation for consumption-related emotions.

² This difference between predictions and current emotions at wave two also remained significant accounting for current emotions at wave one as covariate, $F(1, 85)=5.58$, $p=.02$.

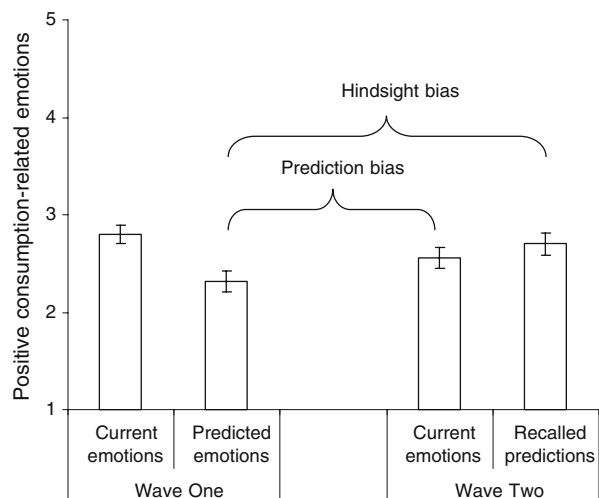
Table 2 Current emotions, predictions, and recalled predictions, by group and country

Country	Group	Wave one		Wave two	
		Current emotions (<i>M</i> (SD))	Predicted emotions (<i>M</i> (SD))	Current emotions (<i>M</i> (SD))	Recalled predictions (<i>M</i> (SD))
A	2 weeks (<i>n</i> =17)	2.97 (0.88)	2.44 (0.91)	2.68 (1.05)	2.82 (1.15)
	4 weeks (<i>n</i> =24)	2.59 (0.75)	1.98 (0.88)	2.40 (0.77)	2.51 (0.86)
UK	2 weeks (<i>n</i> =22)	2.99 (0.86)	2.57 (1.15)	2.85 (1.02)	2.85 (1.05)
	4 weeks (<i>n</i> =23)	2.70 (1.06)	2.36 (1.06)	2.37 (1.08)	2.66 (1.12)
Total	Total (<i>n</i> =86)	2.80 (0.89)	2.32 (1.02)	2.56 (0.98)	2.70 (1.03)

4.2 Perceived prediction after experience

Further planned contrasts showed that not only a prediction bias but also a hindsight bias occurred. Recalled predictions ($M=2.70$) were significantly higher than original predictions ($M=2.32$), $F(1, 82)=14.41$, $p<.01$, $d=0.37$. Recalled predictions ($M=2.70$) were even higher than current emotions at wave two ($M=2.56$), $F(1, 82)=5.28$, $p=.03$, $d=0.13$. These results indicate that participants were not able to recall predictions correctly. They believed that they overestimated emotions, therefore, perceived a small prediction bias; however, the actual prediction bias was larger and in the opposite direction. Again, all ratings of emotions were significantly correlated; recalled predictions correlated higher with current emotions at wave two ($r=.86$) than with current emotions at wave one ($r=.60$) and with predictions ($r=.61$). A regression analysis showed that recalled predictions were not only influenced by original predictions ($\beta=0.21$, $p<.01$), but also, and even stronger, by current experienced emotions at wave two ($\beta=0.74$, $p<.01$), $R^2=.76$, $p<.01$.

Fig. 1 Current emotions, predictions, and recalled predictions (total). Error bars indicate standard errors



To summarize, a prediction bias as well as a hindsight bias for consumption-related emotions were found. Participants predicted a decrease in positive emotions over time. Current experienced emotions at wave two were indeed lower than at wave one; however, participants overestimated this decrease. Thus, participants were too pessimistic, predicting lower positive emotions than they actually experienced (prediction bias about change in emotions). Results indicate that adaptation processes play a role in predicting consumption-related emotions. Moreover, recalled predictions did differ from original predictions and were more strongly influenced by actual experienced emotions at wave two than by original predictions. Participants perceived a prediction bias in the opposite direction than it actually occurred (hindsight bias). No differences between the 2 and the 4-week conditions or the two countries could be found.

5 Discussion

Based upon previous literature on affective forecasting, a prediction bias was expected, and results confirmed that consumers were incorrect in predicting the change in consumption-related emotions. Participants who had just bought a pair of shoes overestimated the decrease of positive emotions towards their shoes over time; they therefore underestimated future positive emotions. These results are consistent with the general conclusion of Kahneman and Snell (1992) that people are not able to predict a change in taste. Results indicate that the inability to predict changes not only applies to taste change but also to the change in consumption-related emotions. In addition, it also applies to durable goods—where the same product is used repeatedly over a longer time span—and not only to products like food that are consumed immediately. Furthermore, it also applies to consumers making real purchases, even though it could be expected that the involvement and motivation to predict correctly is higher with real purchases than in a laboratory setting. To summarize, consumers showed a bias in the prediction of change: although they were able to predict the direction of change, they mispredicted its magnitude. However, even such a gradual error leads consumers to mispredict the overall pleasure derived from consumption over a product's lifetime and may contribute to suboptimal decisions.

One plausible explanation for this prediction bias about change in emotions is an incorrect intuitive theory of adaptation (Gilbert et al. 1998; Loewenstein and Schkade 1999; Wilson and Gilbert 2003). It therefore was examined if intuitive theories about adaptation can be linked to the prediction bias. Results showed that adaptation processes play a role in the predicted decline of emotions and the prediction bias. Participants predicting that they would get used to the shoes in a short period of time also predicted a decline of emotions. Participants misestimating how strongly they would get used to the shoes also misestimated emotions at wave two. These results support the assumption that consumers are holding an intuitive theory of adaptation. They believe that they will get used to the shoes and, therefore, positive emotions will decline over time. However, if consumers overestimate this adaptation, they overestimate the decrease of emotions and consequently underestimate future positive emotions.

The second explanation was that the temporal distance between prediction and experience influences the magnitude of the prediction bias. No support for this argument of temporal separation (MacInnis et al. 2005) could be found in the present study. A plausible explanation seems that the intuitive theory of adaptation is applied on a general level and, therefore, consumers do not differentiate between different points in time. They believe that consumption-related emotions will generally decrease over time but can not exactly estimate this decrease. Therefore, the difference between 2 and 4 weeks might have been too short to show such an effect.

Finally, it was expected that consumers would misremember original predictions (hindsight bias, e.g., Fischhoff 1975). Results showed that indeed a hindsight bias for consumption-related emotions occurred. Recalled predictions were different from original predictions and were more strongly influenced by current experienced emotions at wave two than by original predictions. Participants even perceived a prediction bias in the opposite direction than it actually occurred. Due to the hindsight bias, participants do not realize the prediction bias and are not able to adjust their intuitive theories. This might be a reason why even for frequently bought products—like shoes—no learning can take place (Meyvis et al. 2007).

Some limitations need to be considered. One could argue that shoes can represent completely different meanings and goals for consumers. Some types of shoes have more hedonic value than other types, and for some consumers, shoes in general have more hedonic value than for other consumers. According to Pham (1998), the goal of consumption determines the relative importance of affective versus cognitive considerations in the evaluation of products. Two types of motives can be distinguished in this sense: consummatory or hedonic motives are the basis of consumption behaviors that are intrinsically rewarding, whereas instrumental motives underlie consumption behaviors that are rarely rewarding in themselves and are undertaken to achieve some other goal. In the case of consummatory motives, affective considerations should be more important determinants of behavior and hence perceived as more relevant (as opposed to cognitive considerations) than in the case of instrumental motives, when attention is turned to the product and its performance, generating a more rational, reasoned assessment of performance (Patrick et al. 2007). Thus, affective forecasting is especially relevant when the consumption goal is hedonic, as the goal focuses attention on the self and on feelings, as opposed to the product and its attributes. It would, therefore, be interesting for further studies to take into account differences between participants regarding their consumption goal. Furthermore, our study focuses on one specific product. Further research should examine if the intuitive theory of adaptation and the resulting prediction bias are general phenomena in the consumer context and can be applied to other products. At least the fact that no differences between Austria and the UK were found is an indication that these phenomena are consistent across countries.

The findings suggest several theoretical implications. At first glance, the present results seem inconsistent with the stream of research showing that people tend to systematically overestimate future emotions (impact bias, e.g., Gilbert et al. 1998). As outlined above, in the consumer context, another focus is prevalent. First, the impact bias studies mostly focus on general happiness after an event, whereas in the consumer context, more specific emotions regarding products are central. Second,

the impact bias studies focus on predictions of emotional reactions to a single event, and not on events that extend over time (e.g., durable goods) where the study of a change in emotions is central. Wilson and Gilbert (2003) discuss intuitive theories as one possible cause for the impact bias but argue that such theories would not lead to a systematic overestimation because they are too diverse and differ from individual to individual. In the consumer context, however, not only the focus of research is different to some extent but also it can be assumed that intuitive theories are systematic—brought about by mass media and by advertising. For example, fashion is one domain where a rapid change in taste has become the rule, suggesting that consumers need new products within a few months. Electronic products like mobile phones similarly have a very short product cycle. In such a culture of consumption, it can be assumed that the prevalent intuitive theory among consumers is one of a strong adaptation to products. Consequently, consumers predict that positive emotions about new products decline quickly. These considerations suggest that theoretical explanations about affective forecasting of consumption-related emotions would benefit from the explicit inclusion of two aspects: the specific kind of intuitive theory that consumers hold and the conditions under which these theories are applied. First, the specific intuitive theory needs to be explored in detail. Existing studies show that people believe in adaptation in some domains like noise (Snell et al. 1995), but not in others like disabilities (Ubel et al. 2005). Second, intuitive theories can be more influential in certain situations. For example, Wang et al. (2009) showed that by asking for two predictions instead of one, beliefs about adaptation can be cued and Igou (2004) showed that by priming stability or change, predictions about the time course of emotions can be influenced. Other factors that influence the salience of intuitive theories and their application to predictions need to be explored, for example, involvement or consumption goals.

The results suggest also practical implications for marketers and consumers. First, the role of intuitive theories for consumers' predictions and decisions is relevant for marketers. Because context information can activate certain intuitive theories (Igou 2004), marketing might be able to influence intuitive theories of consumers to make them more accurate. Advertising or specific sale strategies can communicate the intuitive theory of long-lasting emotions associated with a product. In the present study, the intuitive theory of adaptation led consumers to overestimate the decrease of consumption-related emotions; a correction of the intuitive theory would influence consumer decisions. Consumers might for instance be willing to spend more money on products if they believe that positive consumption-related emotions do not fade away quickly. Furthermore, a correction of the intuitive theory would make predictions more accurate, which in turn can enhance consumer satisfaction (Patrick et al. 2007). Second, it is also in the interest of consumers to develop correct intuitive theories in order to make optimal decisions. Intuitive theories do not only influence predictions (Igou 2004) but also recall (Ross 1989). Consumers may, consequently, not only overestimate the decrease of consumption-related emotions in predictions but also exaggerate such a decrease in their recollections. If they believed that the positive emotions associated with the product have decreased over time, they would be more willing to invest in a new product to compensate for the decreased emotions. A correction of the intuitive theory would therefore prevent unnecessary purchases. Advising consumers to correct their intuitive theories would be necessary,

as the hindsight bias prevents consumers from correcting their intuitive theories themselves. This approach of advising consumers in order to avoid decision biases and improve their decision-making is in line with suggestions of Ratner et al. (2008).

To conclude, the present study contributes to the literature on affective forecasting and hindsight bias by focusing on the change in consumption-related emotions of consumers in a field setting. It shows that predictions about change in consumption-related emotions might be incorrect and that an intuitive theory of adaptation is related to mispredictions. The study also points out the relevance of the hindsight bias which hinders consumers from remembering predictions correctly. Therefore, consumers are not able to adjust their intuitive theories; the theories persist, and consumers do not learn from previous prediction errors.

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